



Maszyny cieplne z perspektywy mechaniki kwantowej

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- 1 Cel pracy
- 2 Termodynamika
- 3 Silniki klasyczne
- 4 Warunki endoodwracalne
- 5 Mechanika kwantowa
- 6 Silniki kwantowe



- Przedstawienie teorii termodynamiki kwantowych silników cieplnych
- Wyprowadzenie wielkości termodynamicznych opisujących wybrane silniki kwantowe
- Optymalizacja parametrów silnika

Twierdzenie Carnota

Silnik Carnota wyznacza największą sprawność dla danych temperatur

Twierdzenie Clausiusa

Dla procesu kołowego:

$$\oint \frac{\delta Q}{T} \leq 0$$

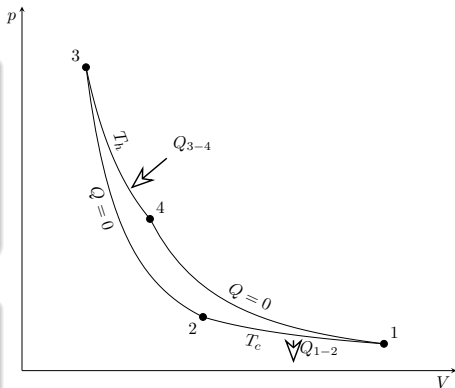
Ciepła

$$Q_{1-2} = nRT_c \ln\left(\frac{V_2}{V_1}\right),$$

$$Q_{3-4} = nRT_h \ln\left(\frac{V_4}{V_3}\right)$$

Sprawność silnika

$$\eta = 1 - \frac{T_c}{T_h}$$



Schemat silnika Carnota

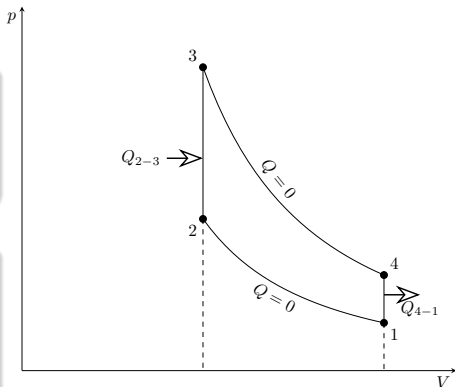
Ciepła

$$Q_{2-3} = nc_V(T_3 - T_2),$$

$$Q_{4-1} = nc_V(T_1 - T_4)$$

Sprawność silnika

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \left(\frac{V_2}{V_1}\right)^{\kappa+1}$$



Schemat silnika Otto

Ciepła

$$Q_{1-2} = nRT_c \ln\left(\frac{V_2}{V_1}\right),$$

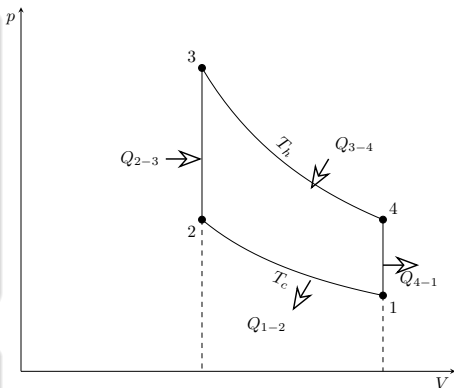
$$Q_{2-3} = nc_V(T_h - T_c),$$

$$Q_{3-4} = nRT_h \ln\left(\frac{V_1}{V_2}\right),$$

$$Q_{4-1} = nc_V(T_c - T_h)$$

Sprawność silnika

$$\eta = 1 - \frac{T_c}{T_h}$$



Schemat silnika Stirlinga

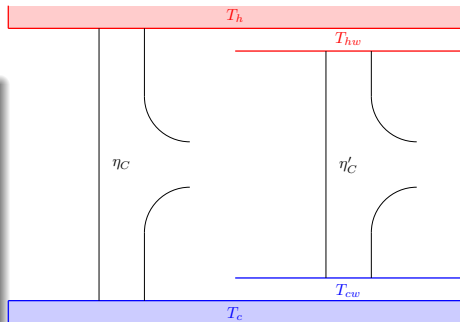
- Równowaga wewnętrzna
- Nierównowaga z otoczeniem

$$x = T_h - T_{hw},$$

$$y = T_{cw} - T_c,$$

$$Q_h \propto x,$$

$$Q_c \propto y$$



Schemat warunków endoodwracalnych

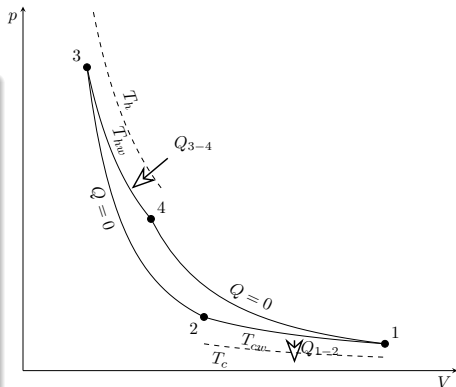
Przemiany izotermiczne

$$Q_{1-2} = \alpha_c \tau_c (T_c - T_{cw}) = -\alpha_c \tau_c y,$$

$$Q_{3-4} = \alpha_h \tau_h (T_h - T_{hw}) = \alpha_h \tau_h x.$$

$$\frac{Q_{1-2}}{T_{cw}} = -\frac{Q_{3-4}}{T_{hw}},$$

$$\frac{\tau_h}{\tau_c} = \frac{\alpha_c y (T_h - x)}{\alpha_h x (T_c + y)}$$



Schemat silnika Carnota

Moc silnika

$$P(x, y) = \frac{\alpha_h \alpha_c x y}{\zeta} \frac{T_h - T_c - x - y}{\alpha_h T_c x + \alpha_c T_h y + (\alpha_h - \alpha_c) x y}$$

Warunki maksimum

$$\begin{aligned} \frac{\partial P}{\partial x} &= 0, & \frac{\partial P}{\partial y} &= 0, \\ \frac{\partial^2 P}{\partial x^2} &< 0, & \frac{\partial^2 P}{\partial x^2} \frac{\partial^2 P}{\partial y^2} - \frac{\partial^2 P}{\partial x \partial y} \frac{\partial^2 P}{\partial y \partial x} &> 0 \end{aligned}$$

Rozwiązanie

$$\frac{x_0}{T_h} = \frac{1 - \sqrt{\frac{T_c}{T_h}}}{1 + \sqrt{\frac{\alpha_h}{\alpha_c}}}, \quad \frac{y_0}{T_c} = \frac{\sqrt{\frac{T_h}{T_c}} - 1}{1 + \sqrt{\frac{\alpha_c}{\alpha_h}}}$$

Sprawność i moc w punkcie maksymalnej mocy

$$\eta = 1 - \frac{T_{cw}}{T_{hw}} = 1 - \sqrt{\frac{T_c}{T_h}},$$
$$P = \frac{\alpha_h \alpha_c}{\zeta} \left(\frac{\sqrt{T_h} - \sqrt{T_c}}{\sqrt{\alpha_h} + \sqrt{\alpha_c}} \right)^2$$

Zmiana temperatury

$$\frac{dT}{dt} = -\alpha_h(T - T_h),$$

$$T(t = 0) = T_2,$$

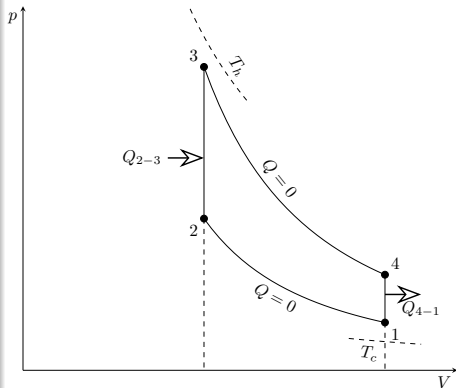
$$T(t = \tau_h) = T_3$$

$$\frac{dT}{dt} = -\alpha_c(T - T_c),$$

$$T(t = 0) = T_4,$$

$$T(t = \tau_h) = T_1$$

$$\frac{T_1}{T_2} = \frac{T_4}{T_3} = \kappa$$



Schemat silnika Otto

Temperatury

$$T_1 = \frac{T_c e^{\alpha_h \tau_h} (e^{\alpha_c \tau_c} - 1) + \kappa T_h (e^{\alpha_h \tau_h} - 1)}{e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1},$$

$$T_2 = \frac{T_c e^{\alpha_h \tau_h} (e^{\alpha_c \tau_c} - 1) + \kappa T_h (e^{\alpha_h \tau_h} - 1)}{\kappa (e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1)},$$

$$T_3 = \frac{T_c (e^{\alpha_c \tau_c} - 1) + \kappa T_h e^{\alpha_c \tau_c} (e^{\alpha_h \tau_h} - 1)}{\kappa (e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1)},$$

$$T_4 = \frac{T_c (e^{\alpha_c \tau_c} - 1) + \kappa T_h e^{\alpha_c \tau_c} (e^{\alpha_h \tau_h} - 1)}{e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1}$$

Ciepła

$$Q_{2-3} = n c_V (T_3 - T_2) = n c_V \frac{(\kappa T_h - T_c)(e^{\alpha_h \tau_h} - 1)(e^{\alpha_c \tau_c} - 1)}{\kappa(e^{\alpha_h \tau_h - \alpha_c \tau_c} - 1)},$$
$$Q_{4-1} = n c_V (T_1 - T_4) = -n c_V \frac{(\kappa T_h - T_c)(e^{\alpha_h \tau_h} - 1)(e^{\alpha_c \tau_c} - 1)}{e^{\alpha_h \tau_h - \alpha_c \tau_c} - 1}$$

Moc

$$P = \frac{n c_V}{\zeta(\tau_h + \tau_c)} \frac{(1 - \kappa)(\kappa T_h - T_c)}{\kappa} \sinh\left(\frac{\alpha_h \tau_h}{2}\right) \sinh\left(\frac{\alpha_c \tau_c}{2}\right) \cosh\left(\frac{\alpha_h \tau_h + \alpha_c \tau_c}{2}\right)$$

Sprawność

$$\eta = 1 - \kappa = 1 - \sqrt{\frac{T_c}{T_h}}$$

Ciepła

$$Q_{1-2} = nRT_{cw} \ln\left(\frac{V_2}{V_1}\right),$$

$$Q_{3-4} = nRT_{hw} \ln\left(\frac{V_1}{V_2}\right)$$

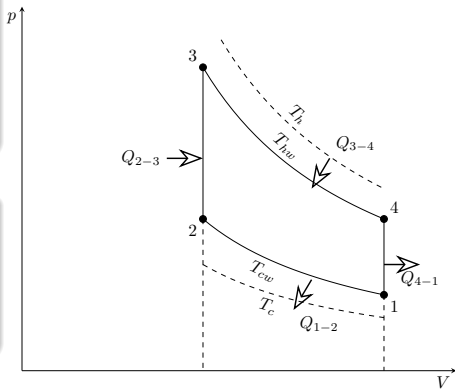
Entropia

$$\frac{Q_{1-2}}{T_{cw}} + \frac{Q_{3-4}}{T_{hw}} = 0,$$

$$\Delta S_{2-3} + \Delta S_{4-1} = 0$$

Sprawność

$$\eta = 1 - \frac{T_{cw}}{T_{hw}} = 1 - \sqrt{\frac{T_c}{T_h}}$$



Schemat silnika Stirlinga

Klasyczny hamiltonian

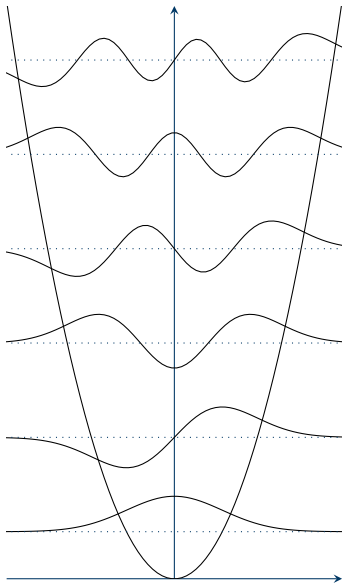
$$\mathcal{H} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$$

Kwantowy hamiltonian

$$\begin{aligned}\hat{\mathcal{H}} &= \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{q}^2 = \\ &= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial q^2} + \frac{1}{2}m\omega^2 q^2\end{aligned}$$

Energia

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$



Zagadnienie własne

$$\hat{S}^2 |sm_s\rangle = \hbar^2 s(s+1) |sm_s\rangle,$$

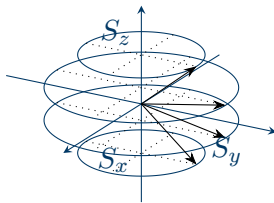
$$\hat{S}_z |sm_s\rangle = \hbar m_s |sm_s\rangle$$

Hamiltonian

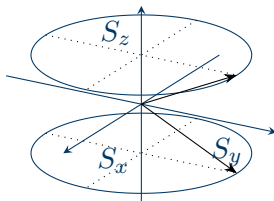
$$\hat{\mathcal{H}} = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = -\gamma \hat{\mathbf{S}} \cdot \mathbf{B}$$

Energia

$$E_{\uparrow} = -\frac{\hbar\omega}{2}, \quad E_{\downarrow} = \frac{\hbar\omega}{2}$$



Stany dla $s = \frac{3}{2}$



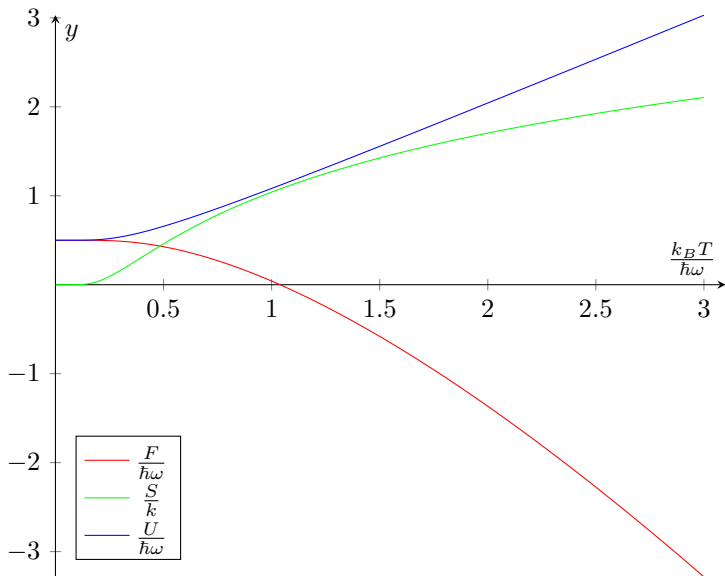
Stany dla $s = \frac{1}{2}$

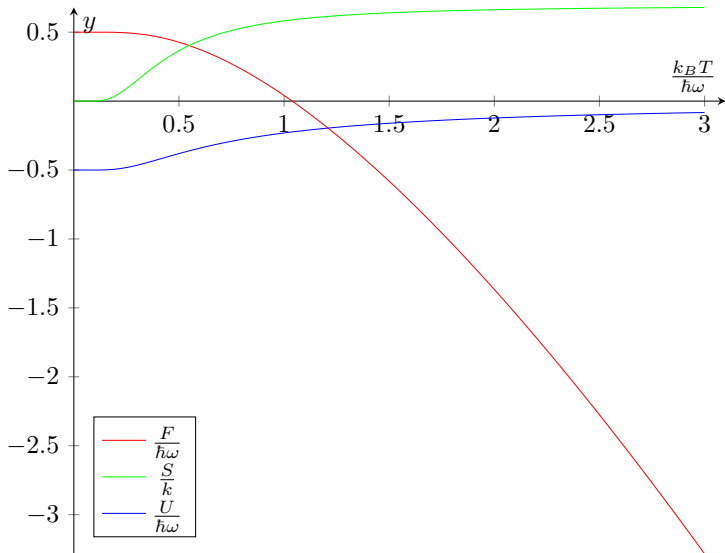
Oscylator harmoniczny

$$\begin{aligned}p_n &= \frac{1}{Z} \exp\left(-\frac{E_n}{k_B T}\right), \\Z &= \frac{\exp\left(-\frac{\hbar\omega}{2k_B T}\right)}{1 - \exp\left(-\frac{\hbar\omega}{2k_B T}\right)}, \\U &= \frac{\hbar\omega}{2} \operatorname{ctgh}\left(\frac{\hbar\omega}{2k_B T}\right), \\S &= \frac{\hbar\omega}{2T} \operatorname{ctgh}\left(\frac{\hbar\omega}{2k_B T}\right) - \\&- k_B \ln\left\{2 \sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right\}\end{aligned}$$

Układ o spinie $\frac{1}{2}$

$$\begin{aligned}p_{m_s} &= \frac{1}{Z} \exp\left(-\frac{E_{m_s}}{k_B T}\right), \\Z &= 2 \cosh\left(\frac{\hbar\omega}{2k_B T}\right), \\U &= -\frac{\hbar\omega}{2} \operatorname{tgh}\left(\frac{\hbar\omega}{2k_B T}\right), \\S &= -\frac{\hbar\omega}{2T} \operatorname{tgh}\left(\frac{\hbar\omega}{2k_B T}\right) + \\&+ k_B \ln\left\{2 \cosh\left(\frac{\hbar\omega}{2k_B T}\right)\right\}\end{aligned}$$





Ciepła przemian

$$Q_{1-2} = 0, \quad Q_{2-3} = \frac{\hbar\omega_2}{2} \left\{ \operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) - \operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right) \right\},$$
$$Q_{3-4} = 0, \quad Q_{4-1} = \frac{\hbar\omega_1}{2} \left\{ \operatorname{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_1}\right) - \operatorname{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_4}\right) \right\}.$$

Praca

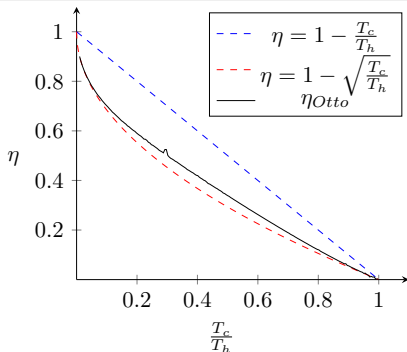
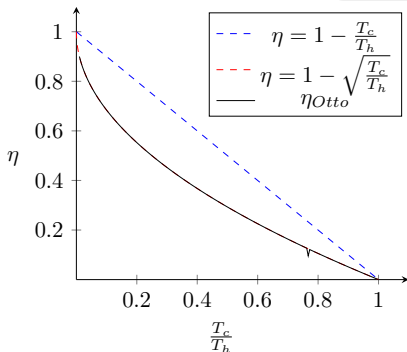
$$-W = \frac{\hbar\omega_2}{2} (1 - \kappa) \left\{ \operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) - \operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right) \right\}.$$

Sprawność

$$\eta = \frac{-W}{Q_{2-3}} = 1 - \kappa = 1 - \frac{\omega_1}{\omega_2}$$

Moc

$$P = \frac{-W}{\zeta(\tau_h + \tau_c)} = \frac{\hbar\omega_2(1 - \kappa)}{2\zeta(\tau_h + \tau_c)} \left[\operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) - \operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right) \right]$$



Ciepła przemian

$$Q_{1-2} = 0, \quad Q_{2-3} = \frac{\hbar\omega_2}{2} \left\{ \operatorname{tgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right) - \operatorname{tgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) \right\},$$
$$Q_{3-4} = 0, \quad Q_{4-1} = \frac{\hbar\omega_1}{2} \left\{ \operatorname{tgh}\left(\frac{\hbar\omega_1}{2k_B T_4}\right) - \operatorname{tgh}\left(\frac{\hbar\omega_1}{2k_B T_1}\right) \right\}.$$

Praca

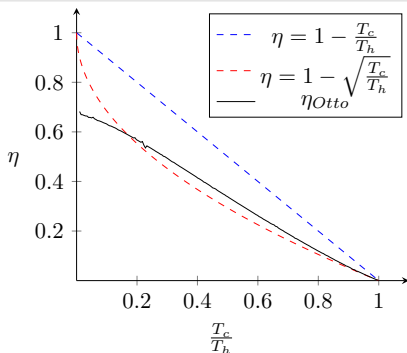
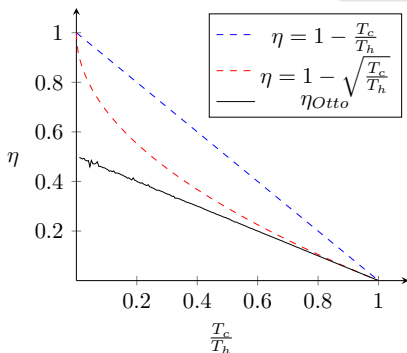
$$-W = \frac{\hbar\omega_2}{2} (1 - \kappa) \left\{ \operatorname{tgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right) - \operatorname{tgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) \right\}.$$

Sprawność

$$\eta = \frac{-W}{Q_{2-3}} = 1 - \kappa = 1 - \frac{\omega_1}{\omega_2}$$

Moc

$$P = \frac{-W}{\zeta(\tau_h + \tau_c)} = \frac{\hbar\omega_2(1 - \kappa)}{2\zeta(\tau_h + \tau_c)} \left[\operatorname{tgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right) - \operatorname{tgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) \right]$$

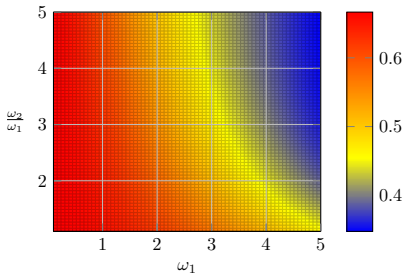
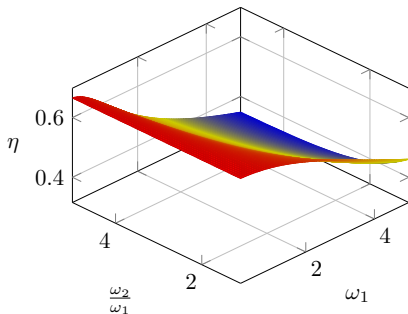


Ciepła przemian

$$\begin{aligned} Q_{1-2} &= k_B T_c \ln \left[2 \sinh \left(\frac{\hbar \omega_1}{2k_B T_c} \right) \right] - k_B T_c \ln \left[2 \sinh \left(\frac{\hbar \omega_2}{2k_B T_c} \right) \right] + \\ &\quad + \frac{\hbar \omega_2}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_2}{2k_B T_c} \right) - \frac{\hbar \omega_1}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_1}{2k_B T_c} \right), \\ Q_{2-3} &= \frac{\hbar \omega_2}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_2}{2k_B T_h} \right) - \frac{\hbar \omega_2}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_2}{2k_B T_c} \right), \\ Q_{3-4} &= k_B T_h \ln \left[2 \sinh \left(\frac{\hbar \omega_2}{2k_B T_h} \right) \right] - k_B T_h \ln \left[2 \sinh \left(\frac{\hbar \omega_1}{2k_B T_h} \right) \right] + \\ &\quad + \frac{\hbar \omega_1}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_1}{2k_B T_h} \right) - \frac{\hbar \omega_2}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_2}{2k_B T_h} \right), \\ Q_{4-1} &= \frac{\hbar \omega_1}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_1}{2k_B T_c} \right) - \frac{\hbar \omega_1}{2} \operatorname{ctgh} \left(\frac{\hbar \omega_1}{2k_B T_h} \right) \end{aligned}$$

Praca

$$-W = k_B T_h \ln \left[\frac{\sinh\left(\frac{\hbar\omega_2}{2k_B T_h}\right)}{\sinh\left(\frac{\hbar\omega_1}{2k_B T_h}\right)} \right] + k_B T_c \ln \left[\frac{\sinh\left(\frac{\hbar\omega_1}{2k_B T_c}\right)}{\sinh\left(\frac{\hbar\omega_2}{2k_B T_c}\right)} \right]$$



Ciepła przemian

$$Q_{1-2} = k_B T_c \ln \left[2 \cosh \left(\frac{\hbar \omega_2}{2k_B T_c} \right) \right] - k_B T_c \ln \left[2 \cosh \left(\frac{\hbar \omega_1}{2k_B T_c} \right) \right] + \\ + \frac{\hbar \omega_1}{2} \operatorname{tgh} \left(\frac{\hbar \omega_1}{2k_B T_c} \right) - \frac{\hbar \omega_2}{2} \operatorname{tgh} \left(\frac{\hbar \omega_2}{2k_B T_c} \right),$$

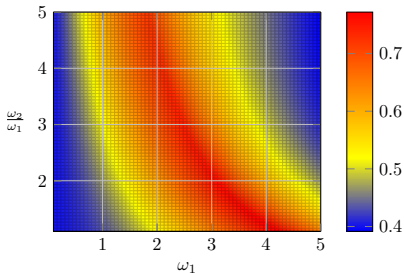
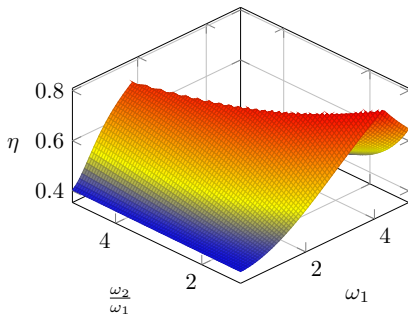
$$Q_{2-3} = \frac{\hbar \omega_2}{2} \operatorname{tgh} \left(\frac{\hbar \omega_2}{2k_B T_c} \right) - \frac{\hbar \omega_2}{2} \operatorname{tgh} \left(\frac{\hbar \omega_2}{2k_B T_h} \right),$$

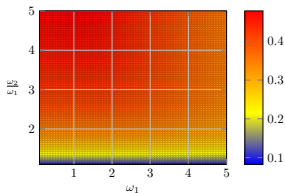
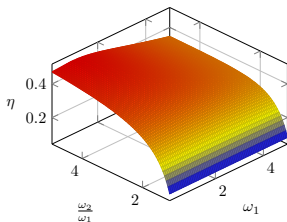
$$Q_{3-4} = k_B T_h \ln \left[2 \cosh \left(\frac{\hbar \omega_1}{2k_B T_h} \right) \right] - k_B T_h \ln \left[2 \cosh \left(\frac{\hbar \omega_2}{2k_B T_h} \right) \right] + \\ + \frac{\hbar \omega_2}{2} \operatorname{tgh} \left(\frac{\hbar \omega_2}{2k_B T_h} \right) - \frac{\hbar \omega_1}{2} \operatorname{tgh} \left(\frac{\hbar \omega_1}{2k_B T_h} \right),$$

$$Q_{4-1} = \frac{\hbar \omega_1}{2} \operatorname{tgh} \left(\frac{\hbar \omega_1}{2k_B T_h} \right) - \frac{\hbar \omega_1}{2} \operatorname{tgh} \left(\frac{\hbar \omega_1}{2k_B T_c} \right)$$

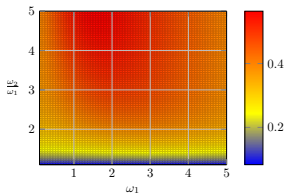
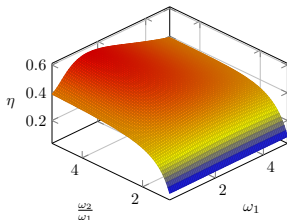
Praca

$$-W = k_B T_h \ln \left[\frac{\cosh\left(\frac{\hbar\omega_1}{2k_B T_h}\right)}{\cosh\left(\frac{\hbar\omega_2}{2k_B T_h}\right)} \right] + k_B T_c \ln \left[\frac{\cosh\left(\frac{\hbar\omega_2}{2k_B T_c}\right)}{\cosh\left(\frac{\hbar\omega_1}{2k_B T_c}\right)} \right]$$





Rysunek: Silnik Stirlinga z oscylatorem harmonicznym



Rysunek: Silnik Stirlinga ze spinem $\frac{1}{2}$